



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

Follow For
More Updates





Class : 12

UNIT TEST-3, NOVEMBER-2024

Register Number

1	2	3	5	9
---	---	---	---	---

BUSINESS MATHEMATICS AND STATISTICS

Time Allowed : 1.30 Hours]

[Max. Marks : 45

PART - I

1. Answer all the questions by choosing the correct answer from the given 4 alternatives **10 x 1 = 10**
2. Write question number, correct option and corresponding answer
3. Each question carries 1 mark

1. Probability which explains x is equal to or less than particular value is classified as
(a) discrete probability (b) cumulative probability (c) marginal probability (d) continuous probability
2. Given $E(X) = 5$ and $E(Y) = -2$, then $E(X - Y)$ is
(a) 3 (b) 5 (c) 7 (d) -2
3. The probability density function $p(x)$ cannot exceed
(a) zero (b) one (c) mean (d) infinity
4. If $x \geq 0$ then $E(x)$ is
(a) < 0 (b) ≥ 0 (c) equal to 0 (d) ≤ 0
5. If $X \sim N(9, 81)$ the standard normal variate Z will be
(a) $Z = \frac{X-81}{9}$ (b) $Z = \frac{X-9}{81}$ (c) $Z = \frac{X-9}{9}$ (d) $Z = \frac{9-X}{9}$
6. If for a binomial distribution $b(n, p)$ mean = 4 and variance = $4/3$, the probability, $P(X \geq 5)$ is equal to :
(a) $(2/3)^6$ (b) $(2/3)^5(1/3)$ (c) $(1/3)^6$ (d) $4(2/3)^6$
7. In a binomial distribution, the probability of success is twice as that of failure. Then out of 4 trials, the probability of no success is
(a) $16/81$ (b) $1/16$ (c) $2/27$ (d) $1/81$
8. The area under the Standard Normal Curve lies between $z = -1$ and $z = 1$ is
(a) 99.74 % (b) 95.44 % (c) 68.26 % (d) 50%
9. A finite subset of statistical individuals in a population is called
(a) a sample (b) a population (c) universe (d) census
10. The standard error of sample mean is
(a) $\sigma / \sqrt{2n}$ (b) σ / n (c) $\sigma / \sqrt{n}$ (d) $\sigma^2 / \sqrt{n}$

PART - II

1. Answer any 4 questions
2. Each question carries 2 marks
3. Question number 16 is compulsory

4 x 2 = 8

11. Let X be a continuous random variable with probability density function $f_x(x) = \begin{cases} 2x, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$
Find the expected value of X.
12. The discrete random variable X has the following probability function $P(X = x) = \begin{cases} kx & x = 2, 4, 6 \\ k(x-2) & x = 8 \\ 0 & \text{otherwise} \end{cases}$ where k is a constant. Show that $k = \frac{1}{81}$
13. A fair coin is tossed 6 times. Find the probability that exactly 2 heads occurs.
14. The mortality rate for a certain disease is 7 in 1000. What is the probability for just 2 deaths on account of this disease in a group of 400? Given $(e^{-2.8} = 0.06)$
15. A wholesaler in apples claims that only 4% of the apples supplied by him are defective. A random sample of 600 apples contained 36 defective apples. Calculate the standard error concerning of good apples
16. What is Standard Error?



PART - III

4 x 3 = 12

1. Answer any 4 questions
2. Each question carries 3 marks
3. Question number 22 is compulsory

17. A continuous random variable X has the following probability function

Value of X=x	0	1	2	3	4	5	6	7
P(x)	0	K	2k	2k	3k	k ²	2k ²	7k ² +k

- (i) Find k (ii) Evaluate $p(x < 6)$, $p(x \geq 6)$ and $p(0 < x < 5)$ (iii) If $P(X \leq x) > \frac{1}{2}$, then find the minimum value of x.
18. Let X be a continuous random variable with probability density function $f(x) = \begin{cases} \frac{3}{x^4}, & x \geq 1 \\ 0, & \text{otherwise} \end{cases}$ Find the mean and variance of X.
19. The distribution of the number of road accidents per day in a city is poisson with mean 4. Find the number of days out of 100 days when there will be (i) no accident (ii) atleast 2 accidents and (iii) at most 3 accidents.
20. X is normally distributed with mean 12 and sd 4. Find $P(X \leq 20)$ and $P(0 \leq X \leq 12)$
21. A sample of 100 items, draw from a universe with mean value 4 and S.D 3, has a mean value 6.5. Is the difference in the mean significant?
22. State any 3 merits of Stratified Random Sampling

PART - IV

1. Answer all the questions

3 x 5 = 15

2. Each question carries 5 marks

23. a) The length of time (in minutes) that a certain person speaks on the telephone is found to be random phenomenon, with a probability function specified by the probability density function $f(x)$ as $f(x) = \begin{cases} Ae^{-x/5}, & \text{for } x \geq 0 \\ 0, & \text{otherwise} \end{cases}$
- (a) Find the value of A that makes f(x) a p.d.f.
- (b) What is the probability that the number of minutes that person will talk over the phone is (i) more than 10 minutes, (ii) less than 5 minutes and (iii) between 5 and 10 minutes.
- (OR)
- b) The probability density function of a random variable X is $f(x) = ke^{-|x|}$, $-\infty < x < \infty$. Find the value of k and also find mean and variance for the random variable.
24. a) One fifth percent of the blades produced by a blade manufacturing factory turn out to be defective. The blades are supplied in packets of 10. Use Poisson distribution to calculate the approximate number of packets containing no defective, one defective and two defective blades respectively in a consignment of 1,00,000 packets ($e^{-0.2} = .9802$)
- (OR)
- b) In a distribution 30% of the items are under 50 and 10% are over 86. Find the mean and standard deviation of the distribution.
25. a) A manufacturer of ball pens claims that a certain pen he manufactures has a mean writing life of 400 pages with a standard deviation of 20 pages. A purchasing agent selects a sample of 100 pens and puts them for test. The mean writing life for the sample was 390 pages. Should the purchasing agent reject the manufactures claim at 1% level?
- (OR)
- b) The average score on a nationally administered aptitude test was 76 and the corresponding standard deviation was 8. In order to evaluate a state's education system, the scores of 100 of the state's students were randomly selected. These students had an average score of 72. Test at a significance level of 0.05 if there is a significant difference between the state scores and the national scores.



SATGURU STUDY CENTRE

NEVER STOP LEARNING. NEVER STOP GROWING...
17/8, Poondi Thangammal St, Nagoor Garden, Tondiarpet, Chennai, Tamil Nadu 600081
An MSME-Certified Learning Centre

CH / M / B. Mat 12 / 2